Revision A | 3.3 V signals | Separate 3.3 V display supply | <=100 mm Type-A 80-way FFC

ribbon

Datei: ribbon.kicad_sch

buffer1

Datei: buffer1.kicad_sch

buffer2

Datei: buffer2.kicad_sch

buffer3

Datei: buffer3.kicad_sch

buffer4

Datei: buffer4.kicad_sch

touch

Datei: touch.kicad_sch

headers

Datei: headers.kicad_sch

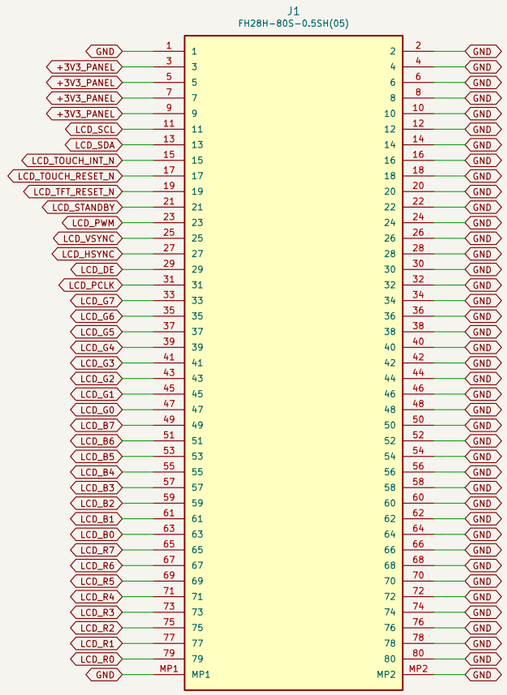
Supplies present

Supplies present

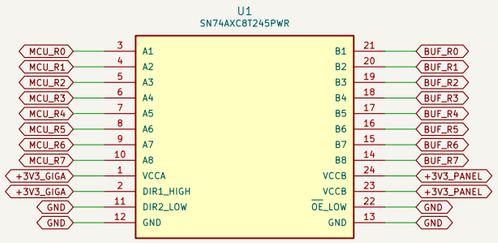
Supplies present

ProtoMate		
Sheet: /		
File: giga-adapter.kicad_sch		
Title: giga-adapter / Smartwin display interface		
Size: A3	Date: 2026-10-14	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 1/8

80-way Type-A (same-side contacts), 0.5mm pitch, 0.30mm mating thickness. GIGA pin n <-> display socket pin 81-n.

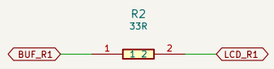

J1
FH28H-80S-0.5SH(05)
GND 1
+3V3_PANEL 3
+3V3_PANEL 5
+3V3_PANEL 7
+3V3_PANEL 9
LCD_SCL 11
LCD_SDA 13
LCD_TOUCH_INT_N 15
LCD_TOUCH_RESET_N 17
LCD_TFT_RESET_N 19
LCD_STANDBY 21
LCD_PWM 23
LCD_VSYNC 25
LCD_HSYNC 27
LCD_DE 29
LCD_PCLK 31
LCD_G7 33
LCD_G6 35
LCD_G5 37
LCD_G4 39
LCD_G3 41
LCD_G2 43
LCD_G1 45
LCD_G0 47
LCD_B7 49
LCD_B6 51
LCD_B5 53
LCD_B4 55
LCD_B3 57
LCD_B2 59
LCD_B1 61
LCD_B0 63
LCD_R7 65
LCD_R6 67
LCD_R5 69
LCD_R4 71
LCD_R3 73
LCD_R2 75
LCD_R1 77
LCD_R0 79
GND MP1
1 1
3 3
5 5
7 7
9 9
11 11
13 13
15 15
17 17
19 19
21 21
23 23
25 25
27 27
29 29
31 31
33 33
35 35
37 37
39 39
41 41
43 43
45 45
47 47
49 49
51 51
53 53
55 55
57 57
59 59
61 61
63 63
65 65
67 67
69 69
71 71
73 73
75 75
77 77
79 79
MP1 MP1
2 2
4 4
6 6
8 8
10 10
12 12
14 14
16 16
18 18
20 20
22 22
24 24
26 26
28 28
30 30
32 32
34 34
36 36
38 38
40 40
42 42
44 44
46 46
48 48
50 50
52 52
54 54
56 56
58 58
60 60
62 62
64 64
66 66
68 68
70 70
72 72
74 74
76 76
78 78
80 80
MP2 MP2
GND
GND
GND

DIR1 = HIGH; DIR2 = LOW; OE = LOW. 33R series termination; 47k input bias. All channels A -> B.

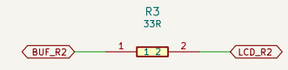




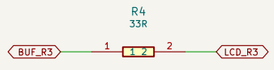
















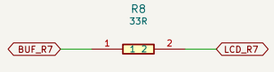




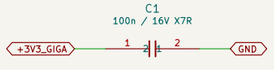


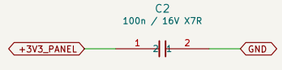












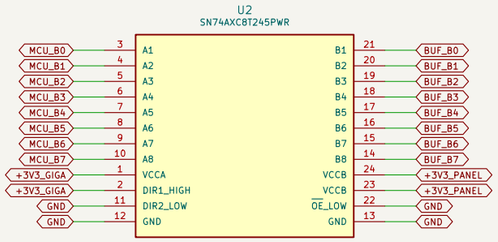
ProtoMate

Sheet: /buffer1/
File: buffer1.kicad_sch

Title: giga-adapter / buffer1

Size: A3	Date: 2026-10-14	**Rev: A**
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Same isolation and termination as U1. Both supplies must be valid for transmission.





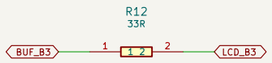




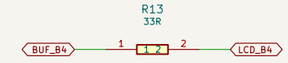




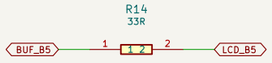












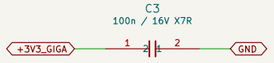


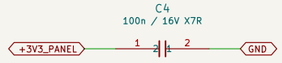












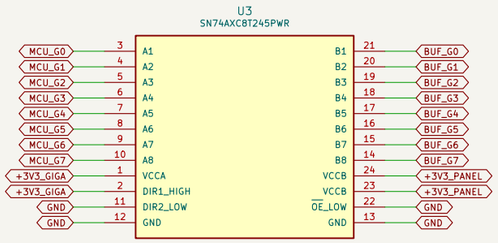
ProtoMate

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File: buffer2.kicad_sch

Title: giga-adapter / buffer2

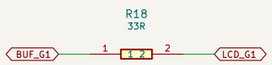
Size: A3	Date: 2026-10-14	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 4/8

RGB24 physical interface. All eight green bits are connected.













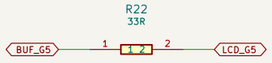














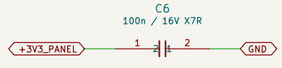












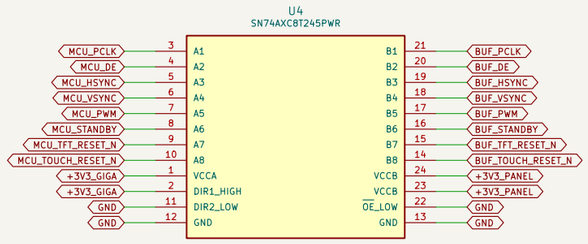
ProtoMate

Sheet: /buffer3/
File: buffer3.kicad_sch

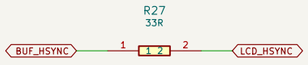
Title: giga-adapter / buffer3

Size: A3	Date: 2026-10-14	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 5/8

PCLK, DE, HSYNC, VSYNC, PWM, STANDBY and active-low resets. PCLK buffered by the same family as RGB.











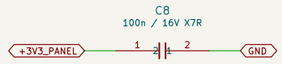




























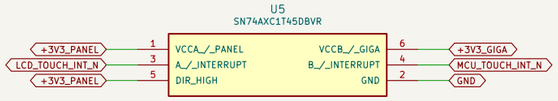
ProtoMate

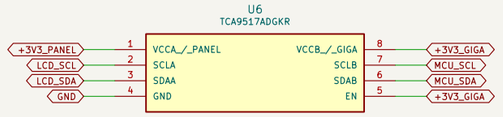
Sheet: /buffer4/
File: buffer4.kicad_sch

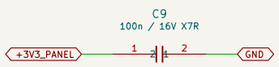
Title: giga-adapter / buffer4

Size: A3	Date: 2026-10-14	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 6/8

TCA9517 A = panel; B = GIGA. GIGA R26/R27 already provide 5.1k pull-ups. AXC1 direction A -> B.



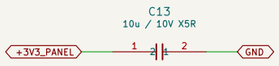


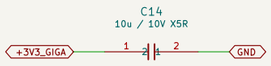


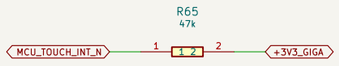






















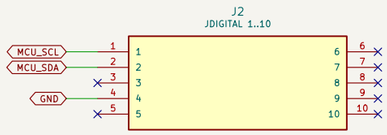
ProtoMate

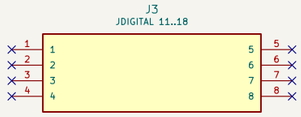
Sheet: /touch/
File: touch.kicad_sch

Title: giga-adapter / touch

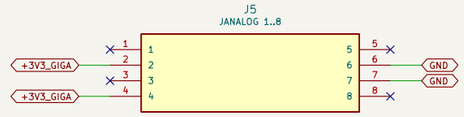
Size: A3	Date: 2026-10-14	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 7/8

NC means no adapter trace. Socket still passes that GIGA contact straight through. Do not populate other header variants.



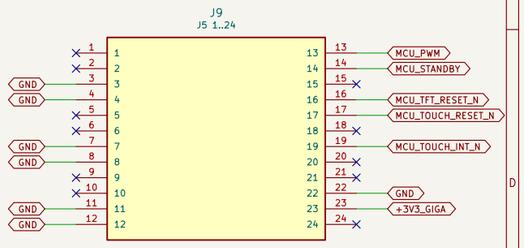












ProtoMate

Sheet: /headers/
File: headers.kicad_sch

Title: giga–adapter / headers

Size: A3	Date: 2026–10–14	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 8/8